Module 4(a) – Negative feedback

Negative Feedback: Four Types of Negative Feedback, VCVS Voltage gain, Other VCVS Equations, ICVS Amplifier, VCIS Amplifier, ICIS Amplifier (No Mathematical Derivation).

Introduction

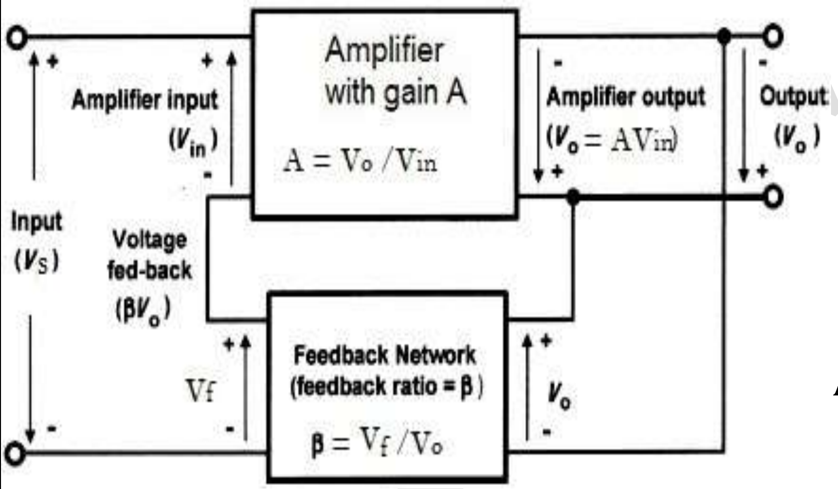
Feedback plays an important role in almost all electronic circuits and is invariably used in amplifiers to improve its performance and to make it more ideal. A feedback amplifier is also called as a closed loop amplifier, since feedback forms a closed loop between the input and the output. In the process of feedback, a part of the output (which can be current or voltage) is sampled and fed back to the input of the amplifier. Hence at the input, there are two signals: input signal and the part of the output which is fed back to the input (feedback signal).Both these signals may be in phase or out of phase.
Hence there are two categories of feedback as – (i) Positive feedback (ii) Negative feedback

Negative Feedback in Amplifiers

Q1. What is Negative feedback in amplifiers? Explain the features of Negative feedback.
In a closed loop amplifier, when the input signal and feedback signal are out of phase (feedback signal is subtracted from input signal), then the feedback is known as Negative feedback.
Features of Negative Feedback (Properties of Negative Feedback)
(i) Desensitize the gain – make gain less sensitive to variations (ii) Decreased gain (ii) Increased bandwidth (iii) Reduction in non-linear operation (iv) Control the input and output resistances (v) Improvement in noise immunity (vi) Make the amplifier system stable

General Feedback Structure

Q2.With a neat block diagram, explain the general feedback structure
Many practical amplifiers use negative feedback in order to precisely control the gain, reduce distortion and improve bandwidth. Figure shows the block diagram of an amplifier stage with negative feedback applied.



The open gain of the amplifier without feedback is A and can be expressed as

$$A = \frac{V_{out}}{V_{in}} \quad \text{or} \quad V_{out} = A\,V_{in}$$

Feedback voltage $V_f = \beta\, V_{out}$
Input $V_S = V_{in} + V_f$
The overall gain of the amplifier with feedback is A_f or $A_{V(CL)}$ can be expressed as

$$\text{The overall gain } A_f = \frac{V_{out}}{V_s} = \frac{A\,V_{in}}{V_{in} + V_f}$$

$$A_f = \frac{A\,V_{in}}{V_{in} + \beta V_{out}} = \frac{A\,V_{in}}{V_{in} + \beta A V_{in}} = \frac{A}{1 + A\beta}$$

Hence, the overall gain with negative feedback applied will be less than the gain without feedback.
For A >>1: ➔ $A_f = (1 / \beta)$. The product term $A\beta$ is known as **Loop gain** of a feedback amplifier and term $(1+ A\beta)$ is called as Amount of feedback.

Example1 : (a) An amplifier with negative feedback applied has an open-loop voltage gain of 50, and (1/10) of its output is fed back to the input . Determine the overall voltage gain with negative feedback applied. (b)If the amplifier's open-loop voltage gain increases by 20%, determine the percentage increase in overall voltage gain.
Solution – Given A = 50; β = 0.1
(a) With negative feedback applied the overall voltage gain will be given by:

$$A_f = \frac{A}{1 + A\beta} = \frac{50}{1 + 50\left(\frac{1}{10}\right)} = 8.33$$

(b) If the amplifier's open-loop voltage gain increases by 20%, The new value of open loop voltage gain is $A_{new} = A + 0.2A = 1.2A = 1.2(50) = 60$

The new overall voltage gain with negative feedback

$$A_{f(new)} = \frac{A_1}{1 + A_1\beta} = \frac{60}{1 + 60\left(\frac{1}{10}\right)} = 8.57$$

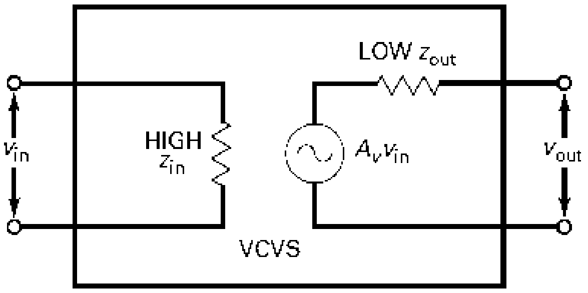
$$\text{The increase in overall voltage gain, expressed as percentage} = \left(\frac{8.57 - 8.33}{8.33}\right) \times 100 = 2.88\%$$

The Four Basic Feedback Topologies and their Comparison

Q. Briefly explain the four types of Negative Feedback topologies.

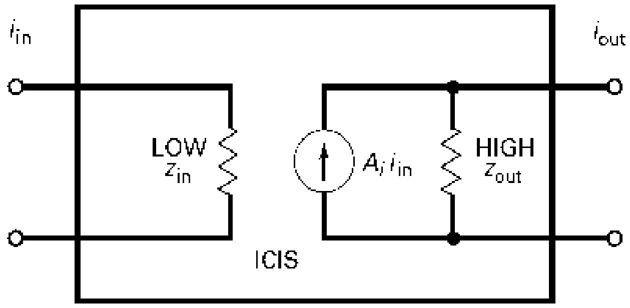
Based on the quantity to be amplified (voltage or current) & on the desired form of output (voltage or current), feedback amplifiers can be classified into four categories

(i) Voltage Controlled Voltage Source (VCVS) amplifier (ii) Voltage Controlled Current Source (VCIS) Amplifiers (iii) Current Controlled Current Source (ICIS) Amplifiers and (iv) Current Controlled Voltage Source (ICVS) Amplifiers



Voltage Controlled Voltage Source (VCVS) Amplifier

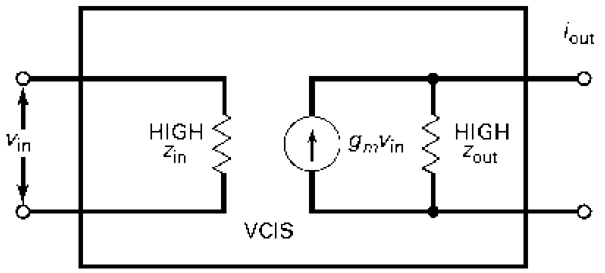
A VCVS is an ideal Voltage amplifier because it has a stabilized voltage gain, infinite input impedance, and zero output impedance



Current Controlled Current Source (ICIS) Amplifiers

A ICIS is an ideal Current amplifier because it has a stabilized current gain, zero input impedance, and infinite output impedance.

Voltage Controlled Current Source (VCIS) Amplifier

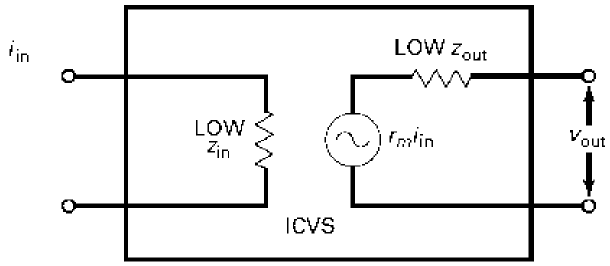


A

VCIS is an ideal Voltage to Current Converter because the input voltage controlling an output current, has high input impedance, and high output impedance

The conversion factor of the VCIS is called Transconductance, symbolized g_m and expressed in Siemens (mhos).

Current Controlled Voltage Source (ICVS) Amplifiers



A

ICVS is an ideal Current to Voltage Converter because the input current controlling an output voltage, has low input impedance, and low output impedance

The conversion factor of the ICVS is called Transresistance, symbolized r_m and expressed in Ohms.

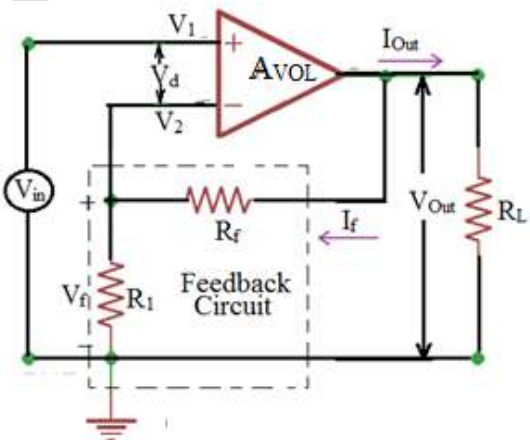
Q. Compare the different negative feedback topologies

Summary Table					**Ideal Negative Feedback**			
Input	Output	Circuit	z_{in}	z_{out}	Converts	Ratio	Symbol	Type of amplifier
V	V	VCVS	∞	0	—	v_{out}/v_{in}	A_v	Voltage amplifier
I	V	ICVS	0	0	i to v	v_{out}/i_{in}	r_m	Transresistance amplifier
V	I	VCIS	∞	∞	v to i	i_{out}/v_{in}	g_m	Transconductance amplifier
I	I	ICIS	0	∞	—	i_{out}/i_{in}	A_i	Current amplifier

Voltage Controlled Voltage Source (VCVS) Amplifier (Voltage Series Feedback Amplifier):

Q. With a neat circuit diagram explain Voltage Controlled Voltage Source (VCVS) Amplifier. Write the expression for closed loop voltage gain, input and output impedances.

Figure shows the circuit diagram of a VCVS Non-inverting amplifier using voltage series feedback with a part of the output voltage fed back in series with the input signal, resulting in an overall gain reduction.



Expression for Closed loop Gain :

Let A_{VOL} be the open loop gain of the amplifier and is related to the output as $V_{out} = A_{VOL} V_d = A_{VOL} (V_1 – V_2)$

From the circuit diagram, $V_1 = V_{in}$ and $V_2 = V_f = \frac{R_1 V_{Out}}{R_1+R_f} --(a)$

Also $V_f = \beta V_{out}$ ------ (b)

From equations (a) and (b) The feedback factor $\beta = \frac{R_1}{R_1+R_f}$

$$V_{Out} = A_{VOL}(V_1 - V_2) = A_{VOL}\left[V_{in} - \frac{R_1 V_{Out}}{R_1+R_f}\right] = A_{VOL} V_{in} - \frac{A_{VOL}\ R_1 V_{Out}}{R_1+R_f}$$

or $$\left[1+\frac{A_{VOL}\ R_1}{R_1+R_f}\right] V_{Out} = A_{VOL} V_{in}$$

Closed loop gain $$A_{V(CL)} = \frac{V_{out}}{V_{in}} = \frac{A_{VOL}}{\left[1+\frac{A_{VOL}\ R_1}{R_1+R_f}\right]} = \frac{A_{VOL}}{[1+A_{VOL}\,\beta]}$$

For an ideal amplifier, A_{VOL} >>1, hence (1+A_{VOL}) = 1 and $A_{V(CL)} = \frac{1}{\beta} = \left(1+\frac{R_f}{R_1}\right)$

Closed loop Input impedance $Z_{in(CL)}$

$Z_{in\,(CL)} = \frac{V_{in}}{I_{in}} = R_{in}(1+A_{VOL}\,\beta)$ where R_{in} is the input resistance without feedback

Input resistance of the amplifier with feedback increases by a factor (1+ $A_{VOL}\,\beta$)

Closed loop Output impedance $Z_{out(CL)}$:

$Z_{out\,(CL)} = \frac{V_O}{I_O} = \left[\frac{R_{Out}}{1+A_{VOL}\,\beta}\right]$ where R_O is the output resistance without feedback

Output resistance of the amplifier with feedback decreases by a factor (1+ $A_{VOL}\,\beta$)

Q. What is Nonlinear distortion and explain how negative feedback reduces non-linear distortion

When the input and output signals are not proportional to each other results in nonlinear distortion which in-turn produces harmonics of the input signal in the output. For instance, if a sinusoidal voltage signal has a frequency of 1 kHz, the distorted output current will contain sinusoidal signals with frequencies of 1, 2, 3 kHz, and so forth. The fundamental frequency is 1 kHz, and all others are harmonics.

The rms value of all the harmonics measured together indicates how much distortion has occurred and is known as **Harmonic distortion**. Harmonic distortion is measured with an instrument called a Distortion analyzer and can be expressed as $THD = \frac{\text{Total Harmonic Voltage}}{\text{Fundamental Voltage}} \times 100\%$

Negative feedback reduces the harmonic distortion and the equation for it is expressed as

Closed loop Harmonic distortion $THD_{CL} = \left[\frac{THD_{OL}}{1 + A_{VOL}\beta}\right]$

where THD_{OL} is open-loop harmonic distortion.

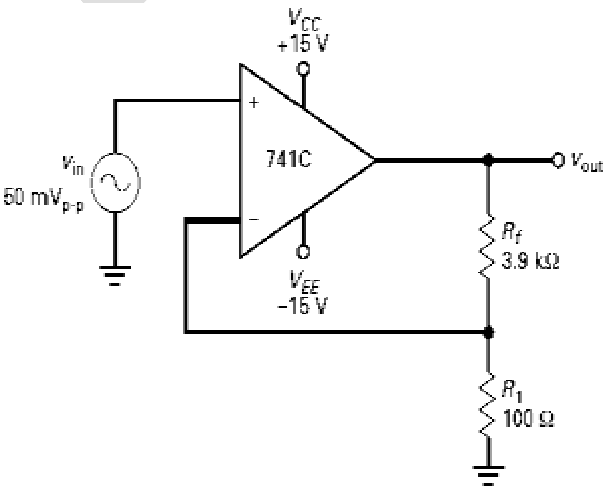
Q. What is gain stability ? Explain.

Gain stability indicates the difference between exact closed loop gain and ideal closed loop gain and expressed as percentage error. The error will be maximum when the open loop gain is minimum and is given by the relation

$$\%\,maximum\ error = \left[\frac{100}{1 + A_{VOL(min)}}\right]$$

Calculate the feedback fraction, the ideal closed-loop voltage gain, the percent error, and the exact closed-loop voltage gain Closed loop input and output impedance for the op-amp circuit shown. Use a typical A_{VOL} of 100,000, R_{in} =2MΩ, R_{CM} = 200 MΩ and R_{out} = 75 Ω for the 741C. Suppose the amplifier has an open-loop total harmonic distortion of 7.5 percent. What is the closed-loop total harmonic distortion?

Solution - The feedback factor $\beta = \frac{R_1}{R_1+R_f} = \frac{100}{3900+100} = 0.025$



Ideal Closed loop voltage gain $A_V = \left(\frac{1}{\beta}\right) = \frac{1}{0.025} = 40$

$$\%Error = \left[\frac{100\%}{1 + A_{VOL}\beta}\right] = \frac{100\%}{1 + 100000 \times 0.025} = 0.04\%$$

Exact Closed loop voltage gain $A_{V(CL)} = \frac{A_{VOL}}{[1 + A_{VOL}\beta]}$

$$A_{V(CL)} = \frac{100000}{1 + 100000 \times 0.025} = 39.084$$

Closed loop input impedance $Z_{in(CL)} = R_{in}(1 + A_{VOL}\beta)$

$$Z_{in(CL)} = 2M(1 + 100000(0.025)) = 5000M\Omega$$

Since $Z_{in(CL)} > 100M\Omega$, consider the value of R_{COM} to calculate $Z_{in(CL)}$ as

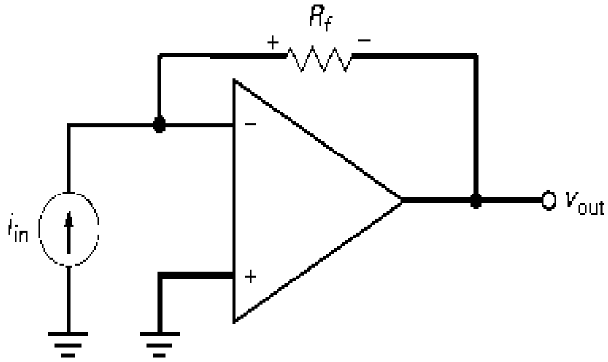
$$Z_{in(CL)} = R_{in}(1 + A_{VOL}\beta)||R_{COM} = (5000M||200M) = 192M\Omega$$

Output resistance with feedback $Z_{out(CL)} = \left[\frac{R_{Out}}{1 + A_{VOL}\beta}\right] = \frac{75}{1 + (100000)(0.025)} = 0.03\Omega$

Closed Loop haormonic Distortion $THD_{(CL)} = \left[\frac{THD_{OL}}{1 + A_{VOL}\beta}\right] = \frac{7.5\%}{1 + 100000(0.025)} = 0.003\%$

Current Controlled Voltage Source Amplifier (Transresistance Amplifier)

Q With a neat circuit diagram explain Current Controlled Voltage Source (ICVS) Amplifier. Write the expression for closed loop voltage gain, input and output impedances.



ICVS amplifier has an input current and an output voltage and is a perfect current-to-voltage converter with zero input impedance and zero output impedance.

The exact output voltage is given by

$$V_{out(CL)} = -A_{V(CL)} V_{in} = -(I_{in} R_f)\left[\frac{A_{VOL}}{1+A_{VOL}}\right] = -I_{in} R_f$$

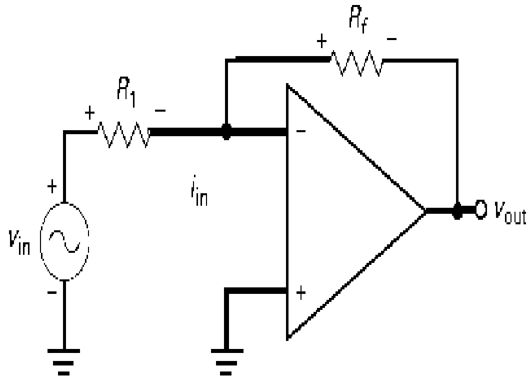
….. where R_f is the transresistance.

Note - Since the inv- input is a virtual ground to voltage not to current; all of the input current must flow through the feedback resistor R_f . Since the left end of this resistor is grounded, the magnitude of the output voltage is given by: $V_{out} = -(i_{in} R_f)$

The closed loop input and output impedances are given by

$$Z_{in(CL)} = \frac{R_f}{1 + A_{VOL}} \quad \text{and} \quad Z_{out(CL)} = \left[\frac{R_{Out}}{1 + A_{VOL}}\right]$$

The Inverting Amplifier

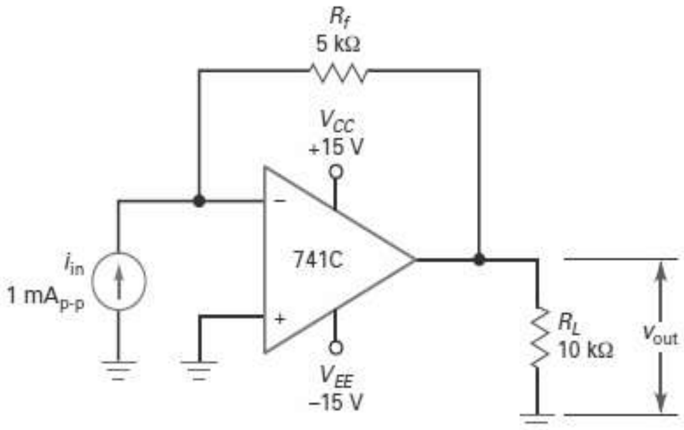


The inverting amplifier is basically a current to voltage converter is shown below. The closed loop gain of the amplifier is given by $A_V = -\left[\frac{R_f}{R_1}\right]$

Because of the virtual ground on the inverting input, the input current equals $I_{in} = \left[\frac{V_{in}}{R_1}\right]$

The output voltage is given by $V_{out} = A_V V_{in} = -\left[\frac{R_f}{R_1}\right](I_{in} R_1) = -I_{in} R_f$

Determine the output voltage, Closed loop input and output impedance for the op-amp



Solution –

$$V_{out} = -I_{in} R_f = -(1 \times 10^{-3})(5 \times 10^{3}) = -5V$$

For 741 op-amp A_{OL} = 100000 and R_{out} = 75 Ω

The closed loop input and output impedances are given by

$$Z_{in(CL)} = \frac{R_f}{1 + A_{VOL}\beta} = \frac{5000}{1 + 100000} = 50\ m\Omega$$

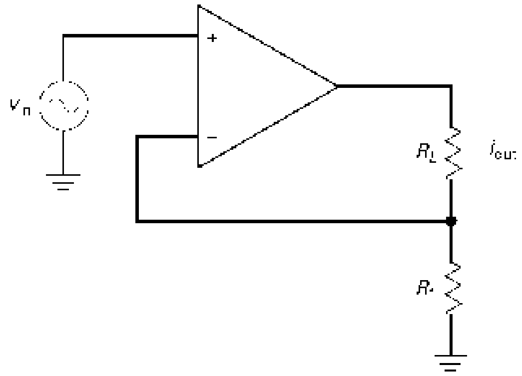
and $$Z_{out(CL)} = \left[\frac{R_{Out}}{1 + A_{VOL}}\right] = \frac{75}{1 + 100000} = 7.5\ \mu\Omega$$

The Voltage Controlled Current Source (VCIS) Amplifier (Transconductance Amplifier)

With a neat circuit diagram explain Voltage Controlled Current Source (VCIS) Amplifier. Write the expression for closed loop voltage gain, input and output impedances.

With a VCIS amplifier, an input voltage controls an output current. Because of the heavy negative feedback in this kind of amplifier, the input voltage is converted to a precise value of output current. The circuit is basically a voltage-to-current converter with very high input impedance and very high output imoegance.

Figure shows a transconductance amplifier with floating load R_L which is acting as the load resistor as well as the feedback resistor. This output current is stabilized; that is, a specific value of input voltage produces a precise value of output current.



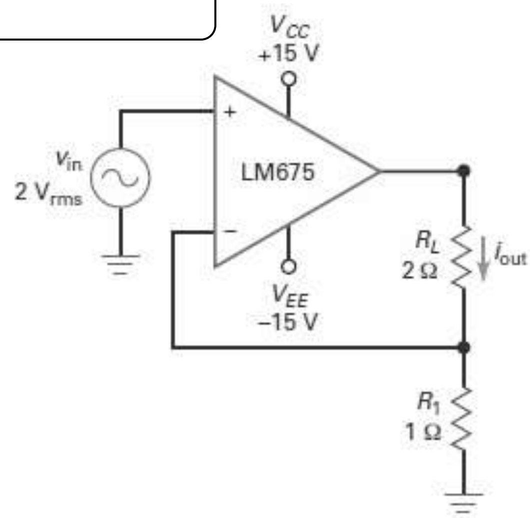
The exact equation for output current is:

$$I_{out} = \left[\frac{V_{in}}{R_1 + \left(\frac{R_1 + R_f}{A_{VOL}}\right)}\right] \cong \frac{V_{in}}{R_1} = g_m\, V_{in} \quad ---- \text{for } R_1 \gg \left(\frac{R_1 + R_f}{A_{VOL}}\right) \text{ and } g_m = \frac{1}{R_1} \text{ is the transconductance}$$

The closed loop input and output impedances are given by

$Z_{in(CL)} = R_{in}[1 + A_{VOL}]$ and $Z_{out(CL)} = R_{out}[1 + A_{VOL}]$

In the given circuit, What is the load current ? The load power? What happens if the load resistance changes to 4Ω ?



Solution - $I_{out(rms)} = \frac{V_{in}}{R_1} = \frac{2}{1} = 2A$

This 2 A flows through the load resistance of 2 Ω, producing a load power:

$P_L = [I_{out}]^2\ R_L = (2)^2(2) = 8$ W

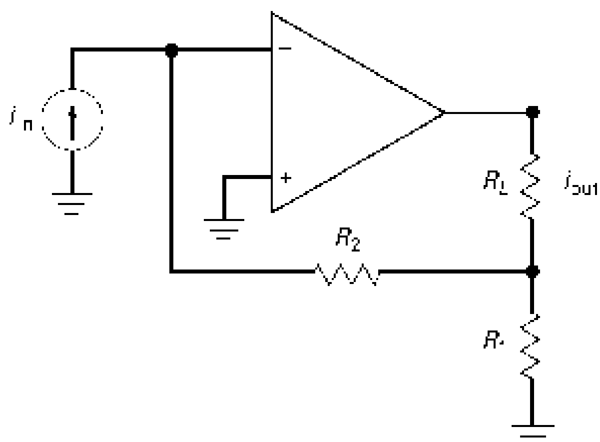
As long as the op amp does not saturate, change in the load resistance to any value does not affect the current and a stabilized output current of 2 A(rms) flows.

The Current Controlled Current Source (ICIS) Amplifier

With a neat circuit diagram explain Current Controlled Current Source (ICIS) Amplifier. Write the expression for closed loop voltage gain, input and output impedances.

An ICIS circuit amplifies the input current. Because of the heavy negative feedback, the ICIS amplifier tends to act like a perfect **current amplifier.** It has a very low input impedance and a very high output impedance.

Figure shows an inverting current amplifier. The closed-loop current gain is stabilized and given by:



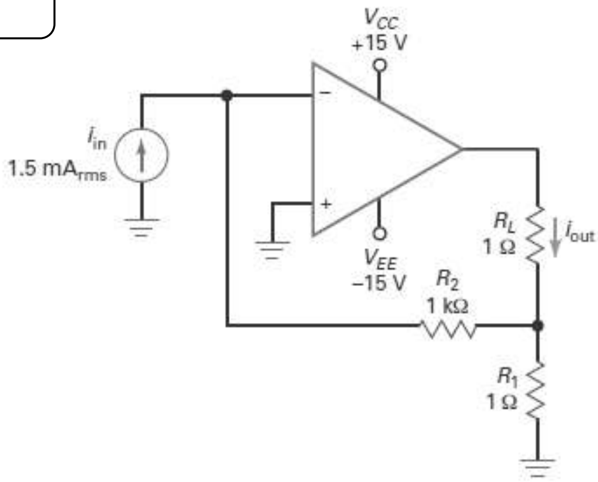
$$A_i = \left[\frac{A_{VOL}(R_1+R_2)}{R_L+A_{VOL}R_1}\right] \cong \left[1+\frac{R_2}{R_1}\right] \quad \text{------- For } A_{VOL}\, R_1 >> R_L$$

The closed loop input and output impedances are given by

$Z_{in(CL)} = \left[\frac{R_2}{1 + A_{VOL}\beta}\right]$ and $Z_{out(CL)} = R_1[1 + A_{VOL}]$

The feedback factor $\beta = \left(\frac{1}{A_i}\right) = \left(\frac{R_1}{R_1 + R_2}\right)$

In the given circuit, What is the load current ? The load power? If the load resistance changes to 2Ω , what is the load current and power?



Solution $A_i = \left[1 + \frac{R_2}{R_1}\right] = 1 + \frac{1000}{1} = 1001$

$I_{out} = A_i I_{in} = 1001(1.5\times10^3) = 1.5$ A (rms)

$P_L = [I_{out}]^2 R_L = (1.5)^2(1) = 2.25$ W

As long as the op amp does not saturate, change in the load resistance to any value does not affect the current and a stabilized output current of 1.5 A(rms) flows.

$P_L = [I_{out}]^2 R_L = (1.5)^2(2) = 4.5$ W

Review Questions

1. Explain the general feedback structure with relevant equations.
2. Mention the important features of Negative feedback.
3. Explain the important properties of Negative feedback used in amplifiers.
4. Explain the classification of feedback amplifiers or Explain the various feedback topologies.
5. Compare Negative feedback amplifier configurations.
6. With a neat circuit diagram explain (i) VCVS amplifier (ii) VCIS amplifier (iii) ICIS amplifier and (iv) ICVS amplifier Write the expression for (a) Closed loop gain (b) input resistance with feedback (c) output resistance with feedback.